

Raj Shah

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EDUCATION

Rutgers University, New Brunswick, NJ | GPA: 3.78 | May 2026

Bachelor of Science | Honors: Computer Science & Data Science | Achievements: Dean's List (2023, 2024, 2025)

Relevant Coursework: Machine Learning, Intro to Artificial Intelligence, Data Structures & Algorithms, Data Visualization

SKILLS & TECHNOLOGIES

Programming: Python, R, SQL, Java

Machine Learning: Scikit-learn, PyTorch, Regression, Classification, Model Evaluation

Data Engineering: ETL/ELT, Data Pipelines, Data Cleaning, Data Visualization, Data Transformation, Batch Processing, Parquet

AI/LLMs: NLP, LangChain, RAG, Transformers (FinBERT)

Data: Pandas, NumPy, Feature Engineering, EDA

Deployment & MLOps: Flask, Docker, GitHub Actions, CI/CD, API Development

Cloud & Tools: AWS, Render, Git, Tableau

PROFESSIONAL EXPERIENCE

AI & ML Engineer - Boundary RSS | Jan 2026 – Present

- Built **end-to-end ML pipeline** classifying lithology across **4 geological datasets**, achieving **~68% accuracy**.
- Standardize multi-source datasets using **Pandas** and **Parquet**, for scalable model **training and inference**.
- Engineered **preprocessing and validation** workflows to improve data quality across dynamic batch prediction inputs.
- Developed batch **inference pipeline** with **predict_proba confidence scoring** for dynamic prediction inputs.
- Design structured **CSV, Parquet** outputs enabling downstream analytics, reservoir modeling, and system compatibility.

Data Analyst Intern - The Marcus Harris Foundation | Aug 2025 – Nov 2025

- Cleaned and validated **1500+ records**, using **SQL**, improving data accuracy and reliability significantly.
- Developed **ETL workflows**, for data transformation, reducing manual reporting time by **25%** overall.
- Performed **exploratory data analysis**, using **Python**, identifying trends and supporting data driven decision making.
- Optimized **SQL queries**, for aggregation and filtering, improving data retrieval efficiency and processing speed.
- Built **Tableau dashboards**, for **100 stakeholders**, improving transparency and data driven decisions significantly.

PROJECTS

Decoding the Fed: NLP-Based Yield Prediction | Jan 2026 – Present

- Built **finance-focused NLP pipeline** using **189 FOMC statements** and **Treasury yield data from 1994-2024**.
- Engineered **FinBERT sentiment, TF-IDF similarity, hawkish/dovish scores** to model Fed communication tone.
- Trained **Random Forest, Lasso, Ridge, Logistic Regression, and XGBoost** across **1, 3, 5-day** yield windows.
- Evaluated models using **RMSE, accuracy, F1 score, and walk-forward validation** to reduce time-series leakage.
- Achieved **0.0865 RMSE** on 2Y prediction and **0.529 accuracy, 0.426 F1** on spread direction.

RU-PATH: AI-Powered Campus Parking & Transit Chatbot | Sept 2025 – Dec 2025

- Built **rule-based RAG chatbot** using **LangChain, DeepSeek API** covering **5 permit types and 6 bus routes**.
- Developed **3 core engines** (chatbot, parking logic, routing) across **1,588 lines** of modular Python code.
- Implemented **semantic search and NLU intent detection**, routing parking vs. bus queries to correct engines.
- Built **Flask REST APIs** enabling real-time multi-turn query handling across **4 Rutgers campuses**.
- Deployed via **Docker, GitHub Actions, Render**, achieving **99% uptime** on production chatbot system.

CNN Digit Classification | Sept 2025 – Dec 2025

- Built **LeNet1 and STN-LeNet architectures** using **PyTorch**, training over **20 epochs** on MNIST dataset.
- Achieved **98.83% test accuracy** on LeNet1, reducing classification error from **2.83% → 1.17%** across training.
- Added **Spatial Transformer Network (STN)** to LeNet, achieving **98.77% best validation accuracy** overall.
- Evaluated both models using **10-class confusion matrix**, identifying per-digit misclassification patterns.
- Visualized training curves and error rates using **Matplotlib, Seaborn** across all **20 training epochs**.